ChatGPT

Performance-Based Advertising Plan for idlookup.ai

Business Context and Goals

idlookup.ai is a subscription service that lets consumers search for information about others and monitor their own digital footprints. The service operates under a business-to-consumer (B2C) model and generates revenue through monthly subscriptions. The **initial six-month lifetime value (LTV)** of a subscriber is approximately **US$18**, so the paid advertising programme must keep the **cost per acquisition (CPA)** below this threshold. Advertising must comply with platform policies and relevant data-privacy regulations – promotions cannot imply stalking or harassment, and cookie and consent banners must be displayed [1]. Clear messaging about privacy and legitimate use builds trust and mitigates policy risk.

Measurement Foundation

1. **Comprehensive tracking:** Implement Google Tag Manager and Microsoft Universal Event Tracking to record page views, sign-ups, subscription purchases, cancellations and key funnel events [1]. Label a **conversion** as a new paid subscription and assign a conversion value equal to the expected LTV (initially US$18) so automated bidding strategies can optimise to value rather than clicks. Capture cancellation dates to refine LTV by cohort.
2. **Data compliance:** Provide cookie banners and privacy notices explaining data collection and honour regional consent requirements. Avoid the use of “free”, “hack”, “stalk” or other sensitive terms [2].
3. **Dashboards and cohort analysis:** Build dashboards that show spend, conversions, CPA, revenue, churn and LTV segmented by channel and cohort. Use cohort tables to compare retention and average revenue per user (ARPU) across acquisition sources [1].

Search Engine Marketing (SEM) Setup

Campaign & Ad-Group Structure

To succeed with long-tail name searches, idlookup.ai should adopt a **granular campaign structure** on both Google and Bing:

Campaign type	Description	Notes
Long-Tail Name Search	Covers queries such as “John Doe public records” or “Jane Smith lookup”. Use Dynamic Search Ads (DSA) or large page feeds to automatically generate ads for thousands of names; group similar names into ad groups and use phrase/exact match to control costs [3].	Long-tail names have low competition and CPC but require automation to manage thousands of variations.

Campaign type	Description	Notes
Topic/Category Search	Targets broader terms like “background check” or “people search”. Use phrase/broad match and DSAs. Bids should be conservative because these keywords tend to be costlier [3].	These terms drive awareness but may require higher CPA thresholds.
Brand & Competitor Campaigns	Protect your own brand terms (“idlookup.ai”) and bid on competitor names to capture intent. Use exact match and high Quality Score; set budgets small and monitor brand CPA.	
Remarketing/ Retargeting	Create campaigns targeting visitors who did not convert. Use RSAs with broad match to recapture high-intent users and employ audience lists for repeat visitors [3].	
Microsoft/Bing Ads	Mirror the Google structure on Microsoft Advertising. Bing’s CPCs are generally lower and its audience skews older; start with manual CPC or Enhanced CPC and transition to Target CPA as data accrues [4]. Use LinkedIn profile targeting (e.g., job roles like IT managers) to reach privacy-conscious professionals.	

Keywords and Bidding

1. **Dynamic Search Ads (DSA):** Use DSAs on both Google and Bing to automatically match long-tail name queries. Build a page feed listing high-value pages and label them so the system focuses on relevant names [5].
2. **Match types:** For name campaigns, rely on **exact and phrase match** to maintain relevance. For topic campaigns, test **broad match** cautiously; negative keywords such as “free”, “hacking”, “stalking” and competitor names prevent unwanted traffic [5].
3. **Initial bidding:** Start with manual CPC bids or Enhanced CPC until each campaign has at least 20–30 conversions. Then activate automated bidding such as **Target CPA** (tCPA). Because the initial LTV is US$18, set a tCPA of about **US$20–$25** for Google and **US$15–$20** for Bing [5] [4]. Adjust tCPA downward as data improves and allocate more budget to campaigns with strong ROAS.
4. **Negative keywords:** Continuously review search-term reports to identify irrelevant or expensive terms. Add negatives to exclude low-intent traffic (e.g., “free background check”, “illegal ways”, “hacker”) [2].

Ad Creative and Extensions

1. **Responsive Search Ads (RSA):** Provide multiple headlines and descriptions so the platform can dynamically test combinations. Use dynamic keyword insertion to insert the searcher’s name while pinning essential copy (e.g., the brand name and CTA). Include benefits such as “Secure & Private Search” and “Unlimited Searches”, and keep messaging trustworthy [6].
2. **Ad extensions:** Add **sitelinks** (e.g., “Name Search”, “Monitor Your Profile”), **callouts** (e.g., “Monthly subscription”, “Cancel anytime”) and **structured snippets** (e.g., “Features: Unlimited searches, Digital footprint monitoring”) [6]. These extensions improve Quality Score and click-through rate.

3. **Testing & AI-generated creative:** Use Google Ads Experiments or Microsoft Experiments to A/B test bidding strategies, match types, ad copy and landing pages. Run tests for at least 2–4 weeks and isolate one variable at a time [6]. An AI agent can generate new headlines/descriptions and summarise asset performance to rotate underperforming assets [7].

Display Network Setup

Responsive Display Ads (RDA)

Create responsive display ads using 5–10 images per aspect ratio. Images should be simple, focusing on trust and privacy (e.g., a person using a computer securely). Provide multiple headlines and descriptions similar to RSA guidelines and include call-to-action assets such as “Start Your Search” or “Protect Your Identity” [8]. Avoid collages or busy backgrounds; the Google Display Network prohibits images that distract from the message.

Audience Strategy

1. **Custom Intent/Segments:** Build custom intent segments using search terms like “people search”, “reverse phone lookup”, “digital footprint monitoring” and competitor brand terms [8]. Test in-market segments such as **“Privacy & Security Enthusiasts”** and affinity segments like **“Online Shoppers”** to find receptive audiences.
2. **Remarketing:** Create remarketing audiences for site visitors who did not convert. Use different time windows (7-day, 30-day) and adjust bids; remarketing CPAs can be higher because returning visitors have greater lifetime value [8].
3. **Exclusion of sensitive categories:** Avoid targeting categories that imply sensitive personal attributes (health, race, political affiliation). Comply with Google’s personalised advertising policies and remove placement on websites related to illegal activities. [8]

Bidding & Budgets

Start with **Maximise Conversions** or low Target CPA on the Display Network and gradually increase budgets as conversion data accumulates [8]. Allocate budgets across custom intent segments and remarketing lists based on CPA and ROAS; bid more aggressively on segments with proven profitability.

Roadmap & Scaling

The advertising programme should evolve over the first year:

Period	Key actions
0–3 months	Launch Google and Bing search campaigns and a small remarketing display campaign. Use manual CPC or low tCPA while gathering baseline data. Implement conversion tracking, UTM parameters, data warehouse and dashboards. Deploy Dynamic Search Ads and at least one A/B experiment. Begin building the AI agent pipeline for data collection and basic alerting [1].

Period	Key actions
3–6 months	Introduce AI-generated creative assets using tools such as Ads Creative Studio or custom LLM prompts. Enable agent-driven budget and bid optimisation. Expand keyword lists based on early winners (e.g., common surnames or cities) and create new ad groups. Launch responsive display campaigns for remarketing and affinity audiences. Run multiple experiments (new landing pages, CTAs) and adjust CPA targets to maintain CPA < LTV [1].
6+ months	Scale budgets on profitable campaigns and consider testing **Performance Max** campaigns, which allow Google's algorithm to allocate spend across search, display, YouTube and Discover. Start exploring affiliate marketing and paid social channels (Meta, LinkedIn) while transferring learnings from search regarding messaging and audience performance. Iterate the AI agent to support cross-channel optimisation and dynamic creative generation at scale [9].

AI Agent-Based Platform Architecture

High-Level Components

1. **Data Ingestion & Storage:** Data connectors pull daily performance metrics from Google Ads, Microsoft Ads and the subscription/billing system. The data ingestion pipeline loads this into a central warehouse (e.g., BigQuery, Redshift) and the transformation layer calculates cost, conversions, CPA, ROAS, retention rate and channel-specific LTV [10].
2. **Monitoring & Alerting:** A monitoring module evaluates metrics against thresholds. For example, it flags campaigns with CPA > 1.25×target (US$22.50 at an initial target of US$18) or conversion volume drops >25 % week-over-week [11]. Alerts are sent to the decision engine for action [12].
3. **Decision Engine:** The core AI agent comprises several submodules:
4. **Budget Optimizer:** Distributes budgets across campaigns based on efficiency (LTV ÷ CPA) and predicted returns [13]. It caps budgets at each campaign's maximum scalable spend and redistributes leftover budget among campaigns needing scale.
5. **Bid Manager:** Chooses bidding strategies (tCPA vs manual CPC) based on conversion volume and adjusts keyword bids proportionally to the deviation of CPA from target [14]. Keywords with CPA above target see reduced bids; those under target have bids raised moderately.
6. **Creative Generator:** Uses large language models to generate new headlines and descriptions according to structured prompts emphasising privacy, legitimate use and a clear CTA [15]. Generated text is filtered for policy compliance and rotated based on performance.
7. **Keyword & Audience Manager:** Mines search-term reports to identify profitable long-tail names, clusters similar queries using NLP and suggests new ad groups or negative keywords [2]. It also proposes custom audiences and remarketing segments.
8. **Experiment Manager:** Schedules and monitors A/B tests or custom experiments. It defines objectives, variables, traffic splits and experiment durations and analyses results for statistical significance [16].
9. **Compliance Monitor:** Scans ads, keywords and targeting settings against Google/Microsoft policies to flag banned content or sensitive targeting [17].
10. **Recommendation & Execution Layer:** The agent compiles recommendations (budget reallocations, bid changes, new creatives, new keywords, experiments) and presents them in a user interface for

human approval. Low-risk changes (e.g., pausing clearly underperforming keywords) may be executed automatically, while higher-impact adjustments require approval [18].

11. **Logging & Audit Trail:** Every decision and action is logged with a timestamp and rationale for transparency and future model training [18].

Data Model and Metrics

Define entities such as **Campaign**, **Ad Group**, **Keyword**, **Audience Segment**, **Creative Asset** and **Experiment**. Each entity stores performance metrics: cost, impressions, clicks, click-through rate (CTR), conversions, CPA, revenue, return on ad spend (ROAS), churn and retention rates [19]. The system also records policy compliance status for each asset.

Workflows

1. **Daily Data Refresh:** At a scheduled time (e.g., 02:00 PST), the system fetches previous-day data, loads it into the warehouse, updates dashboards and runs the monitoring module to check thresholds [20].
2. **Monitoring & Alerting:** Thresholds are defined for each metric; if a campaign's CPA exceeds 1.25×target or conversion volume drops by more than 25 %, an alert is created with context (campaign ID, metrics, timeframe) [11] [12].
3. **Budget Allocation Algorithm:** The agent calculates an efficiency score for each campaign (LTV ÷ CPA) and allocates budget proportionally. Budgets are capped at each campaign's **max scalable spend** and leftover budget is redistributed among campaigns needing scale [13]. See the pseudo-code in the specification for details [21].
4. **Bid Adjustment Logic:** For keywords with ≥5 conversions, if the CPA exceeds the target, bids are decreased proportionally; if the CPA is below target, bids are increased modestly. Keywords with little data see only minor adjustments [14]. The bid manager outputs a list of bid changes and recommended bidding strategies. [22]
5. **Creative Generation & Testing:** The agent uses structured prompts to request five headlines and four descriptions from a language model. Generated assets are filtered for policy compliance and assembled into RSAs or RDAs. Underperforming assets are paused; new assets are generated to replace them [15].
6. **Keyword & Audience Management:** Apply NLP clustering to search-term reports to group similar long-tail names and identify clusters with high conversions and acceptable CPA; create new ad groups for these clusters or add them to the DSA feed [2]. Suggest new in-market or affinity audiences based on behavioural data [16].
7. **Experiment Management:** The experiment scheduler maintains a queue of proposed experiments with defined objectives, variables, control/test groups, dates and traffic splits. Experiments are executed programmatically via the ad platforms and monitored for statistical significance [16].
8. **Compliance Monitoring:** Text and images are scanned for policy violations (e.g., references to illegal activity or sensitive targeting). Ads or targeting settings that violate policies are flagged for review and removed [17].

Implementation Considerations

- **Technology stack:** Use **Python** for ETL, data analysis and decision logic (Pandas, NumPy, Scikit-learn). A cloud data warehouse such as **BigQuery**, **Snowflake** or **Redshift** stores raw and

transformed data; scheduling/orchestration is handled by **Airflow** or **Prefect**. Integrate a large language model (e.g., GPT-4) for creative generation and summarisation [23].
- **Security & privacy:** Access ad platforms via OAuth2 and store tokens securely. Do not target users based on sensitive attributes or use personally identifiable information. Respect GDPR/CCPA requirements and Google/Microsoft data policies [23].
- **Human in the loop:** Provide a UI where marketers can review alerts, recommendations and explanation of decisions. Allow overrides and maintain an audit log for all changes [24].
- **Iterative improvement:** Begin with rule-based heuristics; gradually introduce machine learning techniques (e.g., multi-armed bandits or reinforcement learning) as sufficient data accumulates. Use back-testing to validate new algorithms before deployment [24].

Example Daily and Weekly Flow

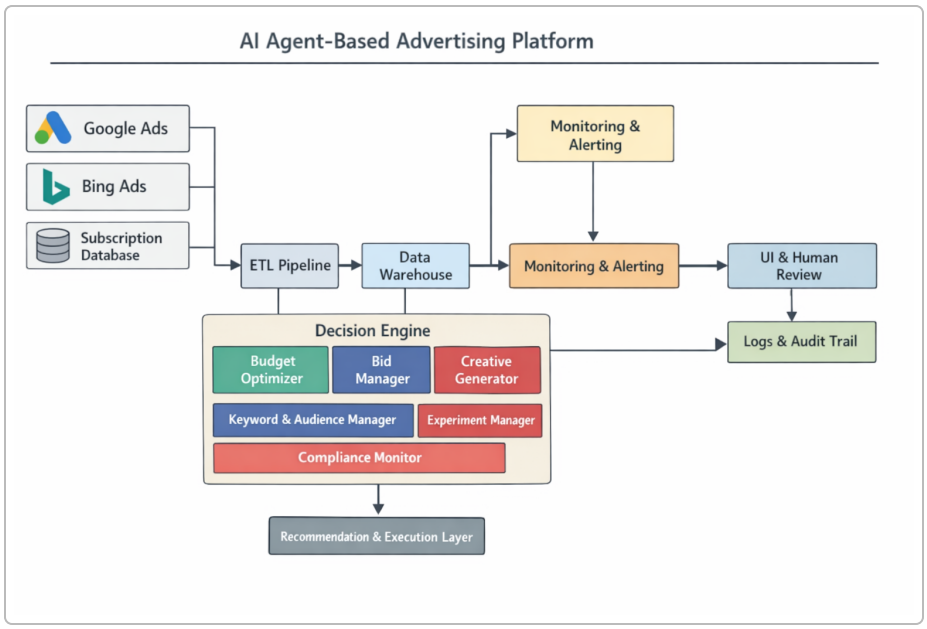
Daily loop: Each morning the agent ingests the prior day's data and discovers that **Campaign A** (name-search) has a CPA of US$36 (target US$25) and conversions have dropped by 30 %. The decision engine recommends reducing Campaign A's budget by 25 %, shifting the freed budget to Campaign B (topic-search) which has CPA = US$18, lowering bids on high-CPA keywords by 40 %, creating an experiment for a new landing page emphasising privacy and adding new long-tail names to the DSA feed [25]. A marketer reviews and approves the changes, and the agent executes them via the API.

Weekly loop: The creative generator produces new seasonal ad copy (e.g., holiday promotions). The experiment manager schedules a two-week A/B test comparing this copy to the existing ads. The compliance monitor verifies that no policies are violated and, upon approval, the test runs. After the test period the agent analyses the results and deploys the winning creative [26].

Process Model and Operating Plan

Operating the AI advertising platform requires a structured process that balances automation with human oversight. The process model below (illustrated in Figure 1) summarises the key stages:

Figure 1. AI Advertising Agent Platform Architecture



1. **Data ingestion & ETL:** Each day, data connectors retrieve performance metrics (impressions, clicks, cost, conversions, revenue, churn) from Google Ads, Microsoft Ads and the subscription database. An ETL pipeline normalises and stores this data in the warehouse.
2. **Monitoring & Alerting:** The system computes KPIs and checks them against thresholds. If anomalies are detected (high CPA, conversion drop, disapproved ads), an alert is created.
3. **Decision Engine:** On receiving an alert or at scheduled intervals, the decision engine analyses data using the Budget Optimizer, Bid Manager, Creative Generator, Keyword & Audience Manager, Experiment Manager and Compliance Monitor. It prepares a set of recommendations.
4. **Human review & execution:** Recommendations are presented in a dashboard. A marketer reviews the rationale and approves or rejects each action. Approved changes are executed via the Google and Microsoft Ads APIs; disapproved changes are logged for training.
5. **Audit & reporting:** All decisions, approvals and actions are stored in an audit log. Weekly and monthly reports summarise performance, ROMI (return on marketing investment), and experiment outcomes. Feedback informs subsequent cycles and algorithm improvements.

Using the Provided Code

The accompanying file `advertising_agent_platform.py` contains a simplified implementation of the AI agent described above. It includes classes to fetch campaign metrics (mocked), compute efficiency scores, allocate budgets, adjust keyword bids, generate prompts for creative generation, cluster search terms and schedule experiments. Developers can extend this framework by integrating the real Google Ads API, Microsoft Advertising SDK, a data warehouse and a language-model API for creative generation. The example `__main__` block demonstrates how to collect metrics, compute budget recommendations and adjust bids.

Conclusion

This plan outlines a comprehensive, AI-driven approach to paid customer acquisition for idlookup.ai. By combining granular campaign structures, data-driven bidding, creative testing, dynamic keyword

management, remarketing and an AI agent that automates day-to-day optimisation, idlookup.ai can achieve efficient growth while keeping CPA below the six-month LTV. The modular architecture supports future expansion into Performance Max, affiliate marketing and paid social channels, and ensures compliance with advertising policies and privacy regulations.

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https://docs.google.com/document/d/1439GAar1JW26fcsyjVk-FGoMyOhjP2ym4l4H-P6RHbA/edit%23heading=h.lsyhi55jprn0

3 4 5 6 7 8 Paid customer acquisition- initial plan and framework - Google Docs
https://docs.google.com/document/d/1439GAar1JW26fcsyjVk-FGoMyOhjP2ym4l4H-P6RHbA/edit%23heading=h.rae9izmu4tne